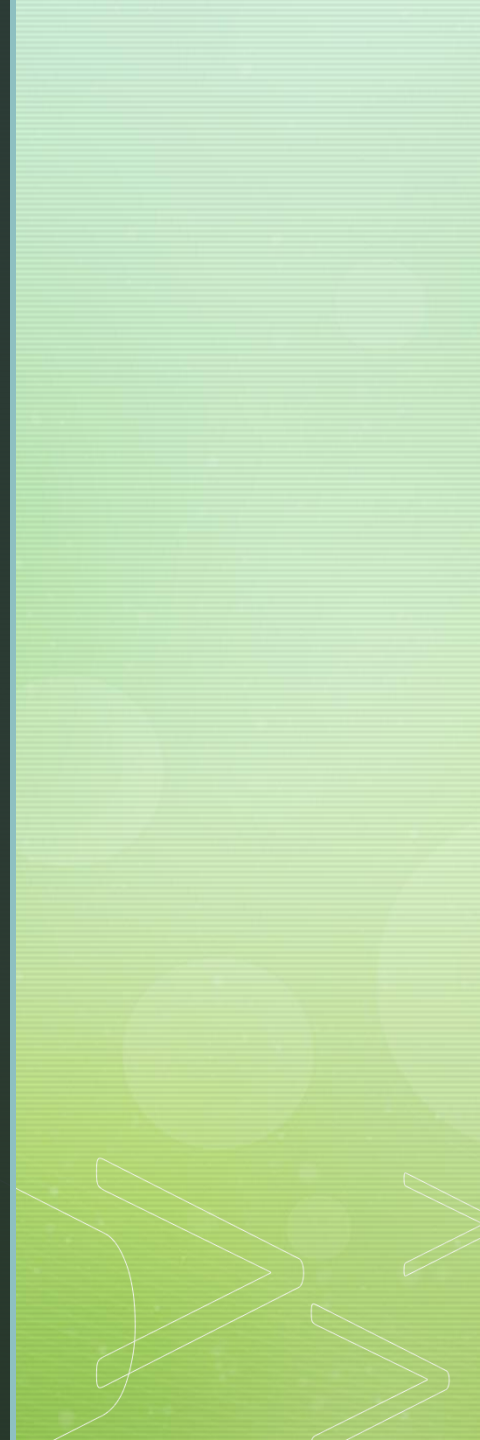




Engineering Ethics Perspective

Intellectual Property and Its Rights

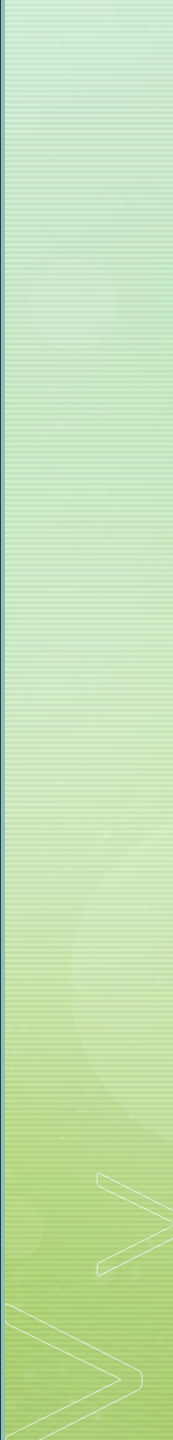


Introduction to Intellectual Property (IP)

- Definition of IP: Creations of the mind, such as inventions, literary and artistic works, designs, symbols, names, and images.
- Importance for engineers and computer scientists.

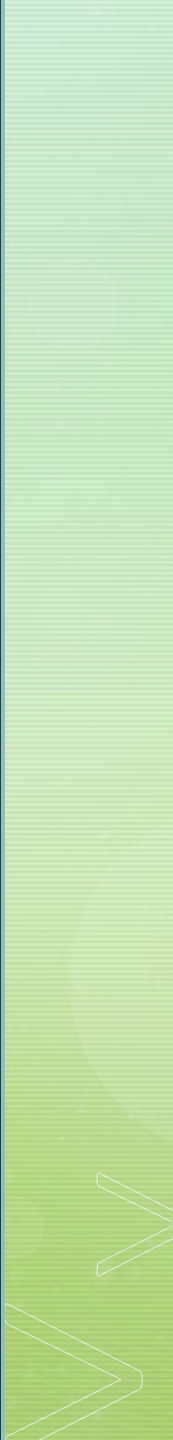


Why IP Matters in Engineering Ethics

- Protects innovation and creativity.
 - Prevents plagiarism and unethical practices.
 - Encourages research and development.
- 

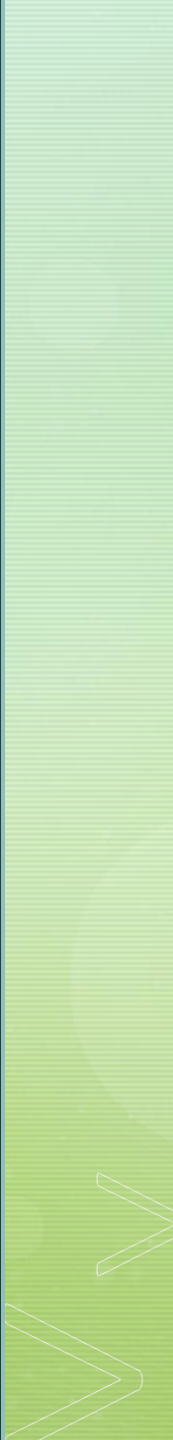


Types of Intellectual Property

- Patents
 - Copyrights
 - Trademarks
 - Trade Secrets
 - Industrial Designs
- 

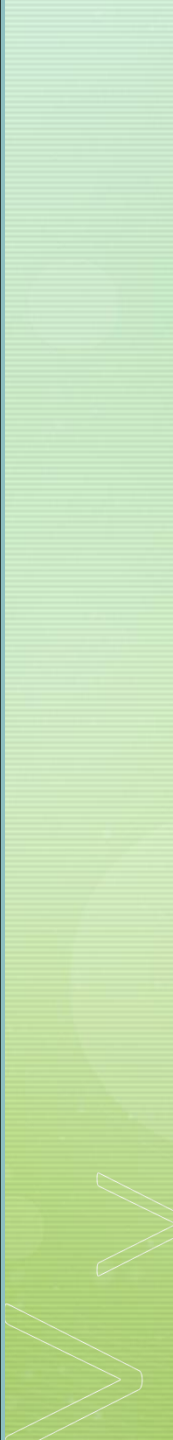


Patents

- Definition: Legal protection for inventions.
 - Duration: Typically 20 years.
 - Examples: New hardware designs, innovative algorithms.
- 



Copyrights

- Definition: Protects literary, musical, and artistic works, including software.
 - Duration: Lifetime of author + 60 years.
 - Examples: Software code, documentation.
- 

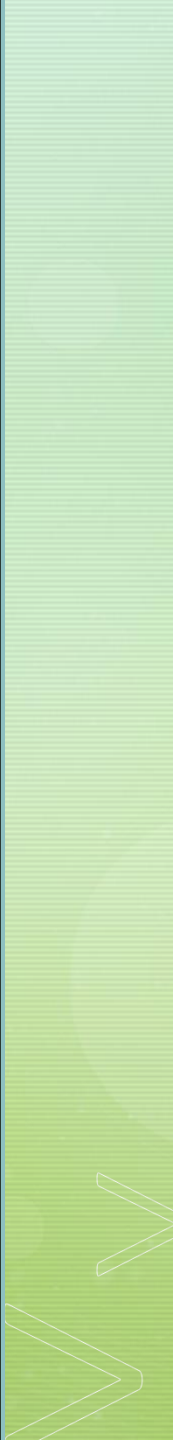


Trademarks

- Definition: Distinguishes goods/services through logos, names, and symbols.
 - Duration: Can be renewed indefinitely.
 - Examples: Tech company logos (Apple, Google).
- 

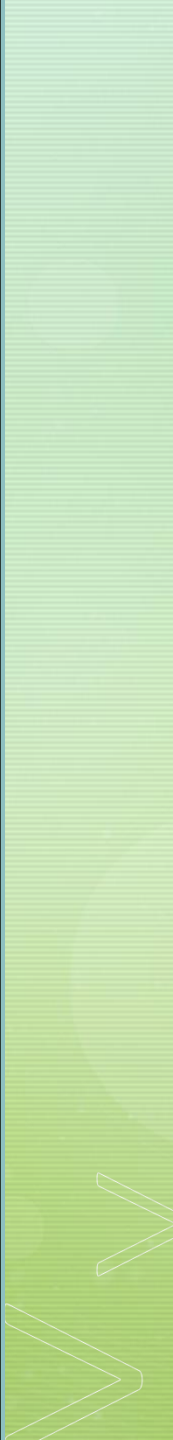


Trade Secrets

- Definition: Confidential business information giving a competitive edge.
 - Examples: Algorithms, formulas (e.g., Google PageRank).
- 



Industrial Designs

- Definition: Aesthetic aspects of an article.
 - Examples: Smartphone design, UI/UX layouts.
- 

IP Rights in Software Development

- Software as copyrightable work.
- Open Source vs Proprietary software.
- Importance of licensing (GPL, MIT).



Case Study: Apple vs Samsung (Patent Dispute)

- Summary: Design patents on smartphones.
- Outcome and ethical implications.



Case Study: Oracle vs Google (Copyright)

- Issue: Use of Java APIs in Android.
- Verdict and impact on software development.

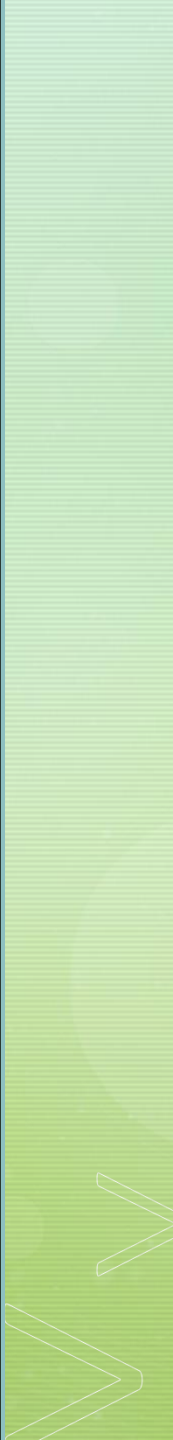


IP Violations & Consequences

- Legal penalties.
- Loss of reputation.
- Impact on careers.

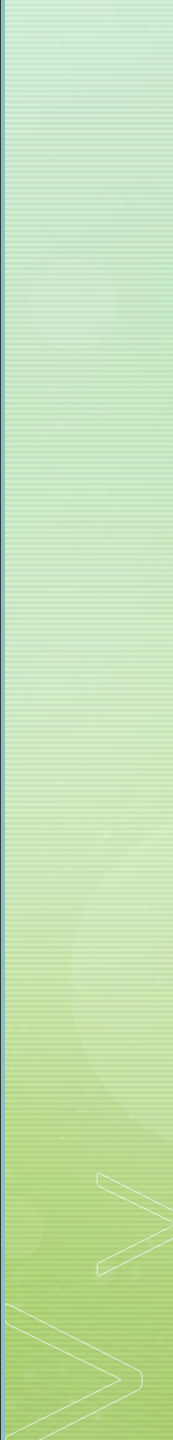


Ethical Issues in IP

- Plagiarism.
 - Reverse engineering.
 - Misuse of open-source licenses.
- 



IP in Academia and Research

- Student projects and plagiarism.
 - Publishing papers and copyright.
 - Collaborative research challenges.
- 



International IP Laws and Treaties

- WIPO (World Intellectual Property Organization).
- TRIPS Agreement.
- International patents.

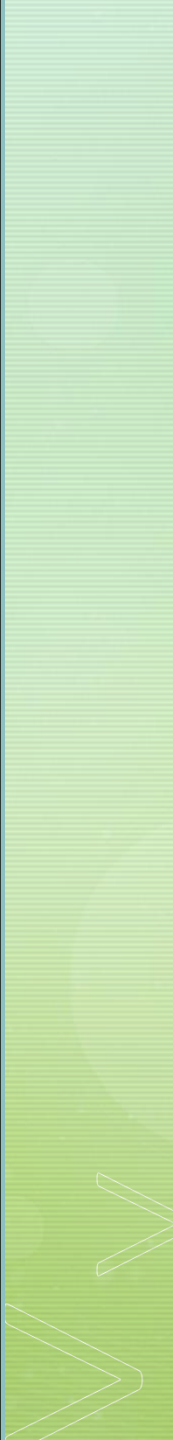


IP and Digital Era Challenges

- Software piracy.
- Digital media sharing.
- AI-generated content issues.

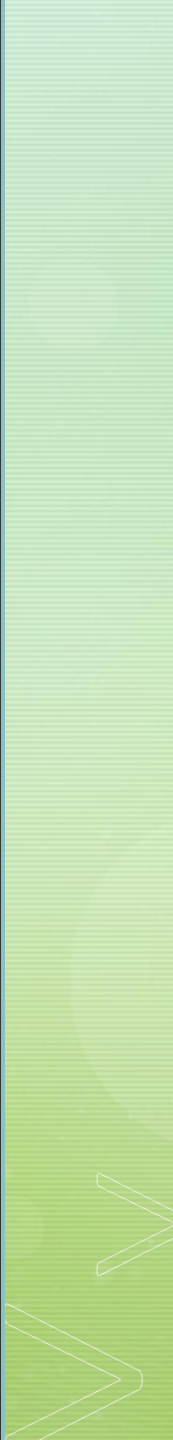


How to Protect Your IP

- Filing patents.
 - Using copyright and trademarks.
 - NDAs for trade secrets.
- 

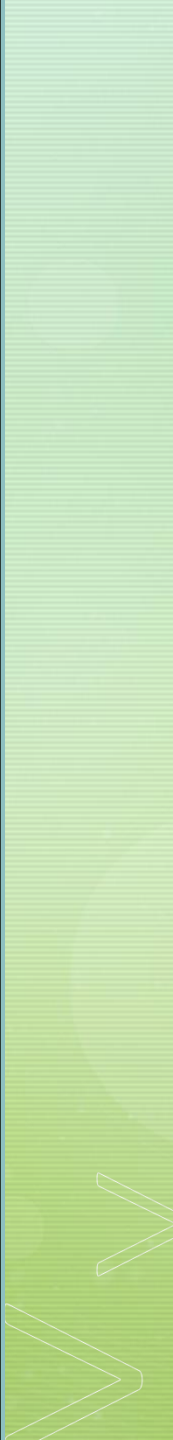


Future Trends in IP

- IP in AI and Machine Learning.
 - Blockchain for IP protection.
 - Global harmonization of IP laws.
- 



Key Takeaways

- IP encourages innovation.
 - Different forms of IP for different creations.
 - Engineers must follow IP laws and ethical practices.
- 

Intellectual Property

The Role of Intellectual Property on Enterprise-led Industry-University-Research (EIUR) Institution Cooperation: Case of Innovation Awards in China

Background

National Innovation System (NIS) framework: shift from government-led to enterprise-led innovation in China.

NIS, RIS, and SIS

- NIS: National Innovation System
- RIS: Regional Innovation System
- SIS: Sectoral Innovation System
- All emphasize multi-actor interactions.

Importance of Enterprises

- Enterprises are the mainstay of innovation.
- Key in activating technological resources and improving efficiency.

Role of IP Strategy

IP protection is vital for technological progress, industrial innovation, and competitiveness.

Research Motivation

Gap: prior studies focus on patents/R&D but overlook commercialization and asymmetric power dynamics in IUR cooperation.

Objectives

- Analyze IPDC policy's effect on E-IUR cooperation.
- Examine enterprise dominance.
- Investigate mediation effects of investment and enforcement.

IPDC Policy Overview

- Launched in 2011 by CNIPA.
- Goal: strengthen IP creation, management, financing, and enforcement.
- 77 cities approved by 2022.

Theoretical Framework

- IPDC enhances cooperation via:
- - Strengthening IP awareness
- - Building service platforms
- - Supporting enterprise leadership.

Research Hypotheses

- H1: IPDC strengthens E-IUR cooperation.
- H2a: Works via enterprise innovation investment.
- H2b: Works via regional IP law enforcement (IPLE).

Data & Sample

- Sample: 3,152 listed firms (2008–2020).
- Sources: WIND, CSMAR, NOSTA, CNIPA.
- ~26,496 firm-year observations.

Methodology

- Approach: Difference-in-Differences (DID) regression.
- Treatment: firms in IPDC cities.
- Outcome: SSTP Awards as measure of deep E-IUR integration.

Variables

- Dependent: E-IUR cooperation (via awards).
- Independent: IPDC policy.
- Controls: Firm age, size, ownership, leverage, ROA, etc.

Results: Main Findings

- IPDC policy significantly strengthens E-IUR cooperation (positive DID coefficient).
- Hypothesis H1 supported.

Robustness Tests

- Parallel trend satisfied.
- PSM-DID (Propensity matching score methods) confirms robustness.
- Endogenous issues controlled with two-stage treatment effects.

Mechanism Analysis

Two mediation channels confirmed:

1. Increased enterprise R&D investment (H2a).
2. Enhanced regional IP enforcement (H2b).

Further Insights

Comparison with IP Demonstration Enterprises (IPDE):

- Stronger effect than IPDC policy.
- Highlights role of enterprise-specific IP initiatives.

Discussion

Key contributions:

- Commercialization focus using awards.
- Asymmetric dependence acknowledged.
- Integration of IP + IUR framework.

Policy Implications

- Government: refine IPDC, promote replication nationwide, improve enforcement.
- Support more cities in IPDC construction.

Conclusion

IPDC policy significantly deepens E-IUR cooperation by stimulating enterprise innovation and strengthening IP law enforcement.

Framed through AIPLA's perspective
(Engineering Ethics & IP)

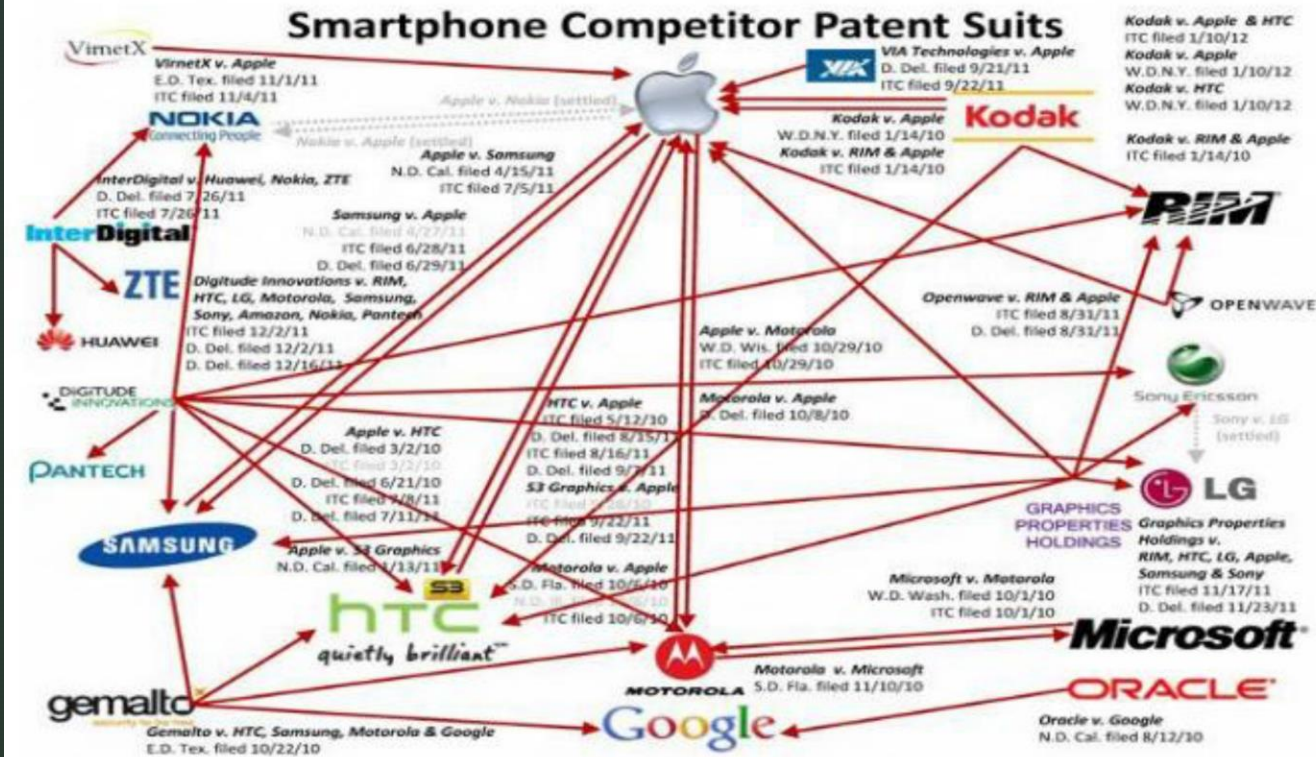


Apple v. Samsung Worldwide Litigation Overview

Background & Scope

- Global smartphone patent wars beginning April 2011
- Claims: design patents, utility patents, trade dress & unfair competition
- Multiple jurisdictions: US, EU (Germany/UK/Netherlands), South Korea, Japan, Australia

"Patent War"



Source: PCMag.com

AIPLA

Parties & Technologies at Issue

- Apple: iPhone/iPad industrial designs (rounded corners, icon grid), UI features
- Samsung: Android smartphones & tablets; SEP and non-SEP counterclaims
- Core themes: innovation, consumer choice, standards, and design value

AIPLA – Who & Why They Weighed In

AIPLA

American Intellectual Property Law Association; filed amicus in Samsung v. Apple (U.S. Supreme Court, 2016).

Purpose

Address the meaning of “article of manufacture” for design-patent damages under 35 U.S.C. §289.

Apple v. Samsung – Apple I: Mobile Phones

PRODUCTS INVOLVED IN THE TRIAL:

iPhone 3G
iPhone 3GS
iPhone 4
iPad 2 3G
iPod Touch

Photos: El Tiempo/Zuma Press (Galaxy S); Apple (iPhone 4)
The Wall Street Journal
Source: WSJ research



Samsung Says...

...THE IPHONE 4 INFRINGES THESE PATENTS:

PATENT '516
Manages the phone's resources and battery by prioritizing data

PATENT '941
Provides a way to package and transmit data more efficiently



PATENT '893

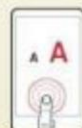
Switching between pictures in a gallery and the camera

Apple Says...

...THE GALAXY S INFRINGES THESE PATENTS:

PATENTS '087, '677
Ornamental design of the iPhone

PATENT '305
Rounded square icons on interface



PATENT '163

Enlarging documents by tapping the screen



Captivate
Continuum
Droid Charge
Epic 4G
Exhibit 4G
Fascinate
Galaxy Ace
Galaxy Prevail
Galaxy S
Galaxy S 4G
Galaxy S II (AT&T)
Galaxy S II (i9100)
Galaxy S II (T-Mobile)
Galaxy S II (Epic 4G)
Galaxy S II (Skyrocket)
Galaxy S Showcase
Galaxy Tab
Galaxy Tab 10.1 (WiFi)
Galaxy Tab 10.1 (4G LTE)
Gem
Indulge
Infuse 4G
Intercept
Mesmerize
Nexus S 4G
Replenish
Transform
Vibrant



PATENT '711

A method allowing multi-tasking while playing music on the phone



PATENT '460

Taking and emailing pictures using a scroll action



PATENT '915

Distinguishes between single-touch and multi-touch gestures



PATENT '381

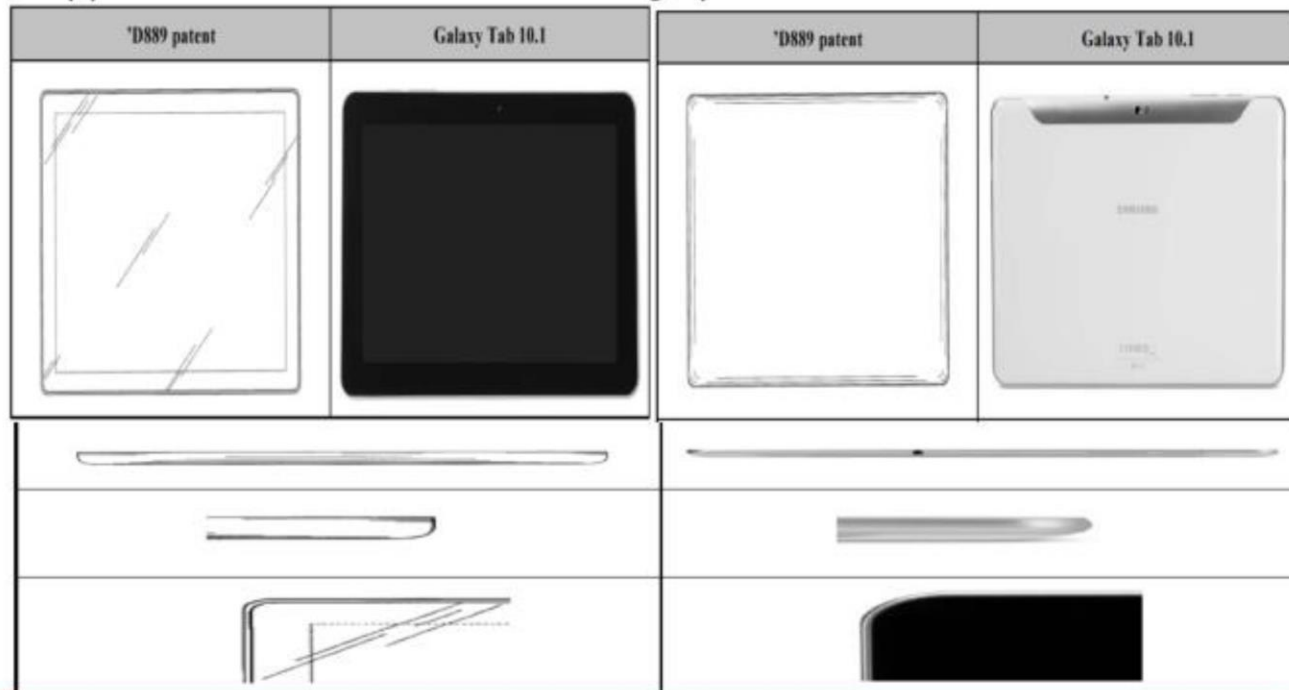
'Bounce-back' feature when scrolling beyond the edge of a page

Source – Prafulla.net

AIPLA

Apple v. Samsung – Apple I: Tablets

- Utility Patents also asserted against Samsung Tablets
- Apple asserts additional tablet design patent



U.S. (N.D. Cal.) – 2012 Jury Verdict

- Jury largely for Apple; initial damages ≈ \$1.05B (later reduced/adjusted)
- Design & utility patent findings; no injunction ultimately changing market
- Set stage for years of appeals and retrials

SCOTUS 2016 – Samsung v. Apple

- Unanimous decision: “article of manufacture” can be a component, not necessarily the whole product (§289)
- Overturned near-\$400M total-profit award and remanded for damages framework
- Key issue: how to identify the relevant article and allocate total profits

AIPLA's Position (2016 Amicus)

- Argued §289 awards “total profit” of the entire article of manufacture covered by the design
- Supported a broad reading grounded in text & history to deter design piracy
- Warned against weakening remedies for design patents

Damages on Remand & 2018 Retrial

- Post-SCOTUS, district court crafted factors for identifying the article of manufacture
- May 2018 jury: \$539M to Apple ($\approx$ \$533M design + \$5.3M utility)
- June 2018: Parties settle U.S. dispute; terms undisclosed

Global Footprint of Litigation

- United States (district court, Federal Circuit, ITC)
- Europe: Germany (injunctions), United Kingdom (no design infringement), Netherlands
- Asia-Pacific: South Korea (split rulings), Japan, Australia

Europe Focus: Germany

- 2011 Düsseldorf court issued EU-wide preliminary injunction on Galaxy Tab 10.1 (later narrowed)
- Subsequent German decisions fluctuated as evidence and designs changed
- Illustrates aggressive use of design rights for fast injunctive relief

Europe Focus: United Kingdom

- 2012 High Court: Samsung's Galaxy Tab did NOT infringe iPad registered design
- Court ordered Apple to publish corrective statements/notices
- Shows divergent design standards & remedies across EU jurisdictions

Asia Focus: South Korea

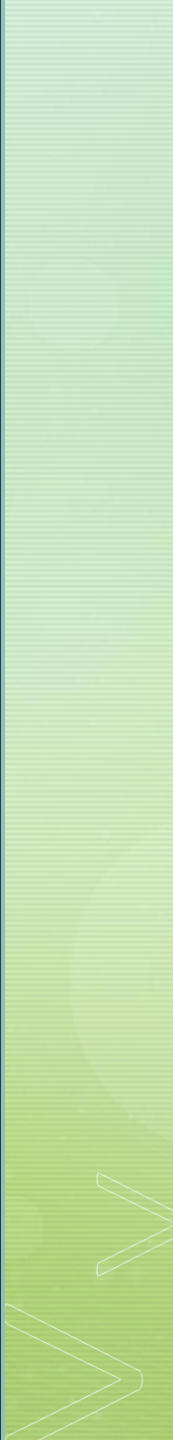
- 2012 Seoul Central District Court: both sides infringed certain patents
- Narrow sales bans on older models; small damages to each party
- Design claims fared differently than in U.S./Germany

Asia-Pacific: Japan & Australia

- Japan: mixed outcomes; Apple lost some claims re multimedia sync/patents
- Australia: early injunction activity and expedited appeals on tablets
- National standards and evidence rules influenced outcomes



Ethics & Policy Themes

- Balancing protection of distinctive designs vs. competition & consumer choice
 - Role of standards-essential patents (FRAND) in countersuits
 - Forum shopping & global compliance burdens for engineers
- 

Design Patents After Samsung v. Apple

- Greater focus on defining the article of manufacture in litigation
- Damages theories scrutinized; risk of windfalls vs. under-deterrence
- In-house teams document component-level design contributions

Practical Takeaways (AIPLA Lens)

- Strong remedies for design patents remain vital against counterfeits
- Clearer tests needed for component vs. whole-product profits
- Early injunctions can shape markets more than damages later

Mini-Timeline: Key Milestones (2011–2018)



Damages Anatomy (Illustrative)

Design Patents (§289)

Total profit on the relevant article of manufacture (SCOTUS clarified could be a component).

Utility Patents (§284)

Reasonable royalty / lost profits; distinct from design-profit remedy.

Engineering Management Checklist

- Document design origins & alternatives; maintain UI/industrial design records
- Assess component vs. device-level exposure in design patents
- Plan for multi-jurisdictional enforcement & compliance

Summary

- SCOTUS, Samsung v. Apple (2016) opinion
- AIPLA Amicus (2016): design damages & article of manufacture
- Reuters coverage of 2018 retrial and settlement; UK/German rulings context